Self-entangled Bellman Equation: From Reinforcement Learning, Least Action to Spectral Self-entangled Superposition States

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Abstract

This research proposes the Self-entangled Bellman Equation by fusing the principle of least action and the classical reinforcement learning Bellman equation, revealing the nature of spectral self-entangled superposition states and solving core problems in classical spectroscopy. The principle of least action governs the evolution of material and optical fields toward global minimal action and minimal energy cost. The classical Bellman equation achieves optimal sequential decision-making for agents. By unifying these two frameworks, the Self-entangled Bellman Equation describes the dynamic evolution of spectral self-entangled superposition states and reveals their intrinsic optimization mechanism as autonomous reinforcement learning agents. Analysis of field spectral data shows that such signals are essentially self-entangled superposition states rather than linear superpositions of classical coherent states. This conclusion explains the Fourier incompleteness in classical spectroscopy and opens a new direction for spectroscopic theory.

Keywords: Self-entangled Bellman equation; principle of least action; reinforcement learning; spectral self-entangled superposition state; Fourier incompleteness

1 Introduction

1.1 Research Background

The principle of least action is a fundamental principle in physics, describing that all physical systems evolve along paths that minimize the action. It provides a unified foundation for classical mechanics, electromagnetism, quantum mechanics, and general relativity. Meanwhile, reinforcement learning (RL) based on the Bellman equation has achieved great success in sequential decision-making tasks.

In spectroscopy, traditional methods based on linear superposition and Fourier transform face severe challenges in complex environments, such as overlapping peaks, matrix effects, and environmental disturbances. Surface-enhanced Raman spectroscopy (SERS) and near-infrared spectroscopy often cannot fully separate components or eliminate nonlinear interference using linear models. These limitations motivate the integration of physical laws and intelligent algorithms.

1.2 Problem Statement

Classical spectroscopy assumes that field spectra are linear superpositions of pure component spectra:

$$S_{\text{class}}(\lambda) = \sum_{k=1}^{N} c_k \varphi_k(\lambda) + \eta(\lambda).$$

However, real-world data universally exhibit *Fourier incompleteness*: linear transforms cannot fully separate overlapping peaks; matrix effects and noise cannot be completely removed; quantitative analysis suffers from systematic errors. These phenomena imply that the linear coherent-state hypothesis is invalid for complex systems.

1.3 Research Objectives

This study unifies the principle of least action and the Bellman equation to establish the Self-entangled Bellman Equation. The goals include:

1. Define field spectra as self-entangled superposition states under self-entangled geometric dynamics;

2. Model spectral evolution as an autonomous reinforcement learning process;
3. Explain the convergence to the optimal steady state under minimal action constraints;
4. Provide a new theoretical framework for spectroscopy and interdisciplinary applications of reinforcement learning in physics.

2 Literature Review

2.1 Theoretical Foundations

The principle of least action originates from Maupertuis and Euler. For a system with Lagrangian L, the action is

$$S = \int_{t_1}^{t_2} L(q, \dot{q}, t)dt,$$

and the physical path satisfies $\delta S = 0$.

The Bellman equation in reinforcement learning is

$$V(s) = \max_a \left[R(s, a) + \gamma V(s')\right],$$

where $V(s)$ is the state value, R is the reward, γ is the discount factor, and s' is the next state.

2.2 Recent Progress

SERS and near-infrared spectroscopy have advanced sensitive detection, but linear models fail in nonlinear coupled systems. AI methods such as deep learning and hybrid quantum-genetic algorithms improve optimization but lack physical interpretability. Quantum information science provides new perspectives but has not been deeply integrated with spectroscopy.

2.3 Research Gaps

Existing literature does not connect reinforcement learning with fundamental optical evolution. Classical spectroscopy cannot explain nonlinear entanglement effects. There is no unified framework for self-optimizing physical fields and intelligent decision-making.

3 Unification of Two Core Principles

3.1 Principle of Least Action

The principle requires that the field evolves to minimize global action. For optical fields, this determines refraction, reflection, interference, and energy cost. In complex mixtures, light–matter interactions produce strong nonlinear geometric entanglement, leading to self-entangled states rather than linear superpositions.

3.2 Classical Bellman Equation

The Bellman equation recursively evaluates optimal policies by maximizing cumulative reward. It decomposes long-term optimization into short-term decision steps, widely used in robotics, energy management, and control systems.

3.3 Unified Framework

Both principles pursue optimality: minimal action vs. maximal value. Mathematically, both involve variational optimization or iterative update. This similarity allows unification. In spectroscopy, the optical field acts as an autonomous agent that optimizes its state toward minimal action, just as an RL agent optimizes value.

4 Core Contradiction: Fourier Incompleteness

4.1 Classical Assumption

Spectra are linear combinations of pure components. Linear tools like Fourier transform and PLSR are used for separation and quantification.

4.2 Phenomena of Fourier Incompleteness

1. Overlapped peaks cannot be fully separated;
2. Matrix effects and environmental drift remain;
3. Systematic errors persist in multicomponent quantification.

4.3 Physical Origin

Fourier incompleteness arises because real spectra are *self-entangled* rather than linearly separable. Nonlocal, nonlinear geometric coupling invalidates the linear superposition assumption.

5 Core Proposition: Self-entangled Superposition States

5.1 Least Action Evolution of Optical Fields

Light evolves to minimize global action and energy cost. The final steady state is stable and measurable, but its internal structure is entangled.

5.2 Definition of Self-entangled Superposition States

A self-entangled superposition state is a high-dimensional Hilbert-space state formed by nonlocal, nonlinear geometric coupling among the optical field, components, environment, and detection system. It cannot be decomposed into independent linear parts.

5.3 Evolution and Steady State

The self-entangled state evolves under the least action constraint until reaching the optimal steady state. Field spectra are exactly this steady state. Linear transforms cannot untangle the nonlinear coupling, causing Fourier incompleteness.

6 Self-entangled Bellman Equation

6.1 Review of the Classical Bellman Equation

$$V(s) = \max_a \left[R(s,a) + \gamma V(s')\right].$$

6.2 Formal Definition of SE-Bellman Equation

The steady-state core equation:

$$\Psi(\lambda) = \mathcal{R}[\Psi|\lambda] + \gamma \cdot \mathcal{U}[\Psi \odot \Psi|\lambda].$$

The dynamic evolution equation:

$$\frac{\partial \Psi}{\partial t} = \mathcal{L}\left(\mathcal{R}[\Psi] + \gamma \mathcal{U}[\Psi \odot \Psi] - \Psi\right),$$

where Ψ is the self-entangled state, $\mathcal{R}$ is the self-reward functional (action cost), $\mathcal{U}$ is the self-entanglement value operator, γ is the entanglement discount factor, and $\mathcal{L}$ is the Laplacian operator.

6.3 Concept Mapping

RL Agent	Self-entangled state $\Psi(\lambda)$
State	Field state at wavelength λ
Action	Self-entanglement operation $\Psi \odot \Psi$
Reward	Self-reward functional $\mathcal{R}$
Value Propagation	Self-entanglement operator $\mathcal{U}$
Discount Factor	Entanglement discount γ

6.4 Equivalence Relation

The least action principle is mathematically equivalent to the optimal convergence condition of the Self-entangled Bellman Equation. The field evolves to minimal action, which corresponds to the RL agent converging to the optimal value function.

7 Theoretical Conclusions

1. Field spectra are **self-entangled superposition states**, not linear coherent superpositions.
2. Optical evolution strictly follows the principle of least action.
3. The Self-entangled Bellman Equation unifies physics and reinforcement learning.
4. Fourier incompleteness arises because linear transforms cannot decompose nonlinear self-entangled coupling.

This framework revolutionizes spectroscopy and provides a rigorous foundation for self-optimizing physical-field intelligence.